

PROGRAM

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#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#define FILENAME "patients.txt"

struct Patient {

    int id;

    char name[50];

    int age;

    char disease[50];

    char doctor[50];

};

// Function to add a patient

void addPatient() {

    struct Patient p;

    FILE *fp = fopen(FILENAME, "a");

    if(fp == NULL) {

        printf("Error opening file!\n");

        return;

    }

    printf("\nEnter Patient ID: ");

    scanf("%d", &p.id);

    printf("Enter Name: ");

    scanf(" %[^\n]", p.name);

    printf("Enter Age: ");

    scanf("%d", &p.age);
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printf("Enter Disease: ");
scanf("%s", p.disease);
printf("Enter Doctor Assigned: ");
scanf("%s", p.doctor);

fprintf(fp, "%d,%s,%d,%s,%s\n", p.id, p.name, p.age, p.disease, p.doctor);
fclose(fp);
printf("Patient added successfully!\n");
}

```

// Function to display all patients

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void displayPatients() {
    struct Patient p;
    FILE *fp = fopen(FILENAME, "r");
    if(fp == NULL) {
        printf("No records found.\n");
        return;
    }

    printf("\n--- Patient Records ---\n");
    while(fscanf(fp, "%d,%49[^\n],%d,%49[^\n],%49[^\n]\n",
        &p.id, p.name, &p.age, p.disease, p.doctor) == 5) {
        printf("ID: %d | Name: %s | Age: %d | Disease: %s | Doctor: %s\n",
            p.id, p.name, p.age, p.disease, p.doctor);
    }
    fclose(fp);
}

```

// Function to search patient by ID

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void searchPatient() {
    struct Patient p;

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int id, found = 0;

FILE *fp = fopen(FILENAME, "r");

if(fp == NULL) {
    printf("No records found.\n");
    return;
}

printf("\nEnter Patient ID to search: ");

scanf("%d", &id);

while(fscanf(fp, "%d,%49[^\n],%d,%49[^\n],%49[^\n]\n",
    &p.id, p.name, &p.age, p.disease, p.doctor) == 5) {
    if(p.id == id) {
        printf("Patient Found: ID: %d | Name: %s | Age: %d | Disease: %s | Doctor: %s\n",
            p.id, p.name, p.age, p.disease, p.doctor);
        found = 1;
        break;
    }
}

if(!found) printf("Patient with ID %d not found.\n", id);

fclose(fp);
}

```

// Function to update patient details

```

void updatePatient() {
    struct Patient p;
    int id, found = 0;
    FILE *fp = fopen(FILENAME, "r");
    FILE *temp = fopen("temp.txt", "w");
    if(fp == NULL || temp == NULL) {
        printf("Error opening file!\n");
    }
}

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        return;
    }

    printf("\nEnter Patient ID to update: ");
    scanf("%d", &id);

    while(fscanf(fp, "%d,%49[^\n],%d,%49[^\n],%49[^\n]\n",
        &p.id, p.name, &p.age, p.disease, p.doctor) == 5) {
        if(p.id == id) {
            printf("Enter new Name: ");
            scanf(" %[^\\n]", p.name);
            printf("Enter new Age: ");
            scanf("%d", &p.age);
            printf("Enter new Disease: ");
            scanf(" %[^\\n]", p.disease);
            printf("Enter new Doctor Assigned: ");
            scanf(" %[^\\n]", p.doctor);
            found = 1;
        }
        fprintf(temp, "%d,%s,%d,%s,%s\\n", p.id, p.name, p.age, p.disease, p.doctor);
    }

    fclose(fp);
    fclose(temp);

    remove(FILENAME);
    rename("temp.txt", FILENAME);

    if(found) printf("Patient updated successfully!\\n");
    else printf("Patient with ID %d not found.\\n", id);
}

```

```

// Function to delete patient
void deletePatient() {
    struct Patient p;
    int id, found = 0;
    FILE *fp = fopen(FILENAME, "r");
    FILE *temp = fopen("temp.txt", "w");
    if(fp == NULL || temp == NULL) {
        printf("Error opening file!\n");
        return;
    }

    printf("\nEnter Patient ID to delete: ");
    scanf("%d", &id);

    while(fscanf(fp, "%d,%49[^\n],%d,%49[^\n],%49[^\n]\n",
        &p.id, p.name, &p.age, p.disease, p.doctor) == 5) {
        if(p.id == id) {
            found = 1;
            continue; // skip writing this record
        }
        fprintf(temp, "%d,%s,%d,%s,%s\n", p.id, p.name, p.age, p.disease, p.doctor);
    }

    fclose(fp);
    fclose(temp);

    remove(FILENAME);
    rename("temp.txt", FILENAME);

    if(found) printf("Patient deleted successfully!\n");
}

```

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    else printf("Patient with ID %d not found.\n", id);
}

int main() {
    int choice;
    do {
        printf("\n--- Hospital Management System ---\n");
        printf("1. Add Patient\n");
        printf("2. Display Patients\n");
        printf("3. Search Patient\n");
        printf("4. Update Patient\n");
        printf("5. Delete Patient\n");
        printf("0. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1: addPatient(); break;
            case 2: displayPatients(); break;
            case 3: searchPatient(); break;
            case 4: updatePatient(); break;
            case 5: deletePatient(); break;
            case 0: printf("Exiting...\n"); break;
            default: printf("Invalid choice!\n");
        }
    } while(choice != 0);

    return 0;
}
```

OUTPUT

--- Hospital Management System ---

1. Add Patient
2. Display Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Exit

Enter your choice: 2

--- Patient Records ---

ID: 101 | Name: Ravi Kumar | Age: 40 | Disease: Fever | Doctor: Dr. Mehta

ID: 102 | Name: Sita Devi | Age: 30 | Disease: Diabetes | Doctor: Dr. Sharma

ID: 103 | Name: Arjun Singh | Age: 25 | Disease: Asthma | Doctor: Dr. Patel

ID: 104 | Name: Meena Joshi | Age: 50 | Disease: Heart Disease | Doctor: Dr. Rao

ID: 105 | Name: Vikram Das | Age: 35 | Disease: Fracture | Doctor: Dr. Nair